

29.9% of returns aren't genuine. They carry **\$4.64M** of refund dollars — and now we know which ones.

A retailer's return-audit team reviews every disputed return the same way — spreadsheet speed, inconsistent criteria. We rebuilt **60,000 orders** into a PySpark/MLlib pipeline that triages each one into legitimate, policy abuse, fraud, or wardrobing, priced what's actually at stake, and separately segmented customers to surface the product-quality problem no fraud label can see.

NON-LEGITIMATE REFUND EXPOSURE

\$4.64M

43.8% of all refund dollars requested, from just **29.9%** of orders — bad actors skew toward higher-value claims

CLASSIFIER PERFORMANCE (TEST SET)

0.999

weighted F1, Random Forest — synthetic-dataset caveat on slide 5, not a production guarantee

HIDDEN IN THE "LEGITIMATE" MAJORITY

38%

of returns in the largest, 96%-legitimate customer segment are still quality-issue driven

60,000 orders · 35 features, Kaggle e-commerce return-abuse dataset · Fraudulent Return claims average **\$434.73** vs. **\$141.14** for legitimate returns — a **3.1x** gap · three independent PySpark MLlib approaches: multi-class classification (Random Forest vs. Logistic Regression), K-Means clustering, and refund-exposure regression · this dataset is synthetic, so treat 0.999 F1 as a pipeline proof, not a production number (more on slide 5).

One review queue, *three different problems* — and the industry already knows it.

A return-audit team currently reviews every disputed return the same way, with no systematic way to tell fraud from policy abuse from a genuine quality defect. This isn't hypothetical: e-commerce carries a structurally higher return burden than brick-and-mortar, and this project sits squarely inside an industry-wide shift already underway.

NRF · 2025 RETAIL RETURNS LANDSCAPE

19.3%

of online sales are returned — above the 15.8% all-channel average. \$849.9B in merchandise returns forecast for 2025.

APPRISS RETAIL x DELOITTE · 2024

15.14%

of all returns are fraudulent — \$103B lost in 2024, across 60+ retailers studied. 60% of execs flag wardrobing specifically; 85% now deploy AI to fight it.

OUR DATASET, BY COMPARISON

17.9%

our Fraudulent Return + Wardrobing rate — close to Appriss's 15.14%. Class **prevalence** is realistic; class **separability** is not (see slide 5).

THE QUESTION THIS PROJECT ANSWERS

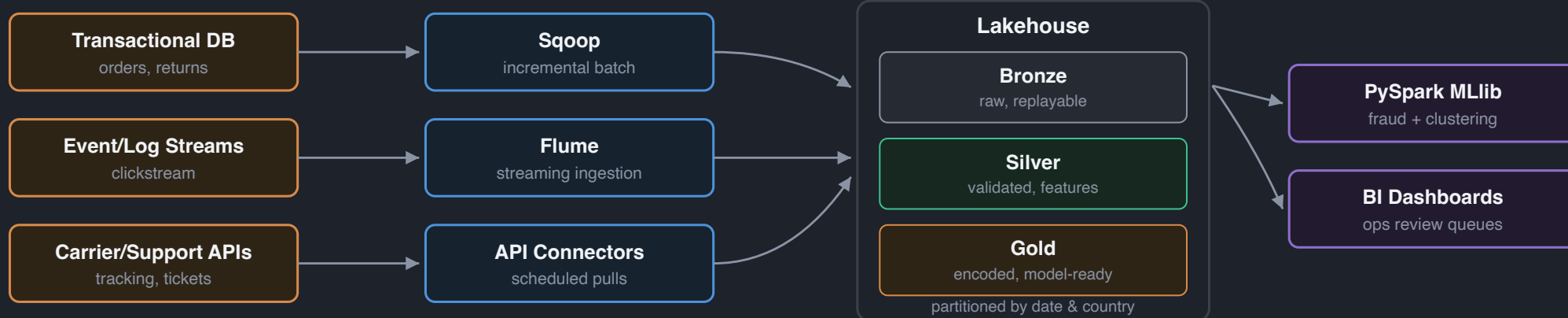
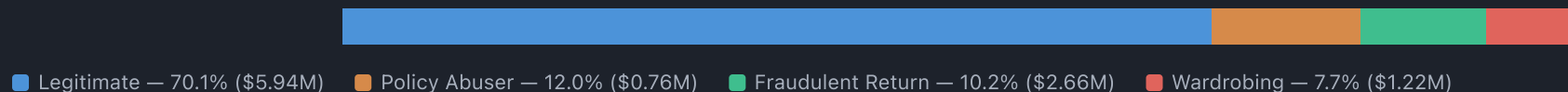
Which returns are fraud, which are policy abuse, and which are genuine quality failures — and what does each one actually cost?

Three sub-problems, one dataset: **return fraud** → classification on abuse_type; **product quality** → clustering cut by category; **return risk** → refund-exposure regression.

WHY THIS IS A TRIAGE PROBLEM, NOT A POLICY PROBLEM

The highest-value lever is routing each return to the right team fast — not tightening policy across the board, which would punish the 70% of customers who did nothing wrong. **Three different interventions, three different owners**, is the recommendation this whole pipeline builds toward.

A realistic imbalance, engineered into a *Bronze→Silver→Gold* pipeline.



VERIFIED, THEN GENERALIZED TO PRODUCTION SCALE

0 missing values, 0 duplicate order_id, 0 logical violations — checked programmatically before anything downstream trusted this data. Bronze keeps a permanent, replayable raw record (explicit `StructType`, not inferred); Silver is cleaned and feature-engineered once, with outliers IQR-capped rather than deleted (extreme values are often the fraud signal itself); Gold is the only layer any model touches. This notebook runs directly on the static CSV — Sqoop/Flume/API ingestion above is the production-scale design the pipeline is built to generalize to.

What the three models actually tell us — *and the risks that travel with them.*

KEY FINDINGS

- **Non-legit returns skew expensive, not just frequent.** 29.9% of orders carry 43.8% of refund dollars (\$4.64M) — fraud claims average 3.1× a legitimate return.
- **The classifier earns its complexity.** Random Forest hits perfect precision/recall where a free 4-flag heuristic tops out at F1 0.61.
- **The quality problem is broad-based, not one vendor.** Quality-issue rates band 31.5%–35.2% across all 12 categories — a systemic pattern, not a vendor hunt.
- **A simpler model wins honestly on refund pricing.** Linear Regression beats both a naive baseline (40% RMSE) and a more complex GBTR regressor.
- **Our dataset's fraud rate is industry-realistic.** 17.9% (Fraud + Wardrobing) sits close to Appriss Retail's reported 15.14% — the scale here is real.

RISKS & FUTURE ROADMAP

- **Risk — synthetic-data overfit.** Treat this model as a pipeline proof-of-concept; re-train on real labeled outcomes before go-live, budgeting for 80–90% F1, not 99.9%.
- **Risk — no production ground-truth labels.** Real retailers rarely have a clean `abuse_type` column — build human-in-the-loop labeling from the review queue this system creates.
- **Risk — the quality finding explains *what*, not *why*.** Broad-based isn't root-caused; that needs a packaging/supplier QA audit this dataset can't run.
- **Roadmap.** Real-time scoring at return-request time, a feedback loop from reviewer overrides back into re-training, and a rolling time-based validation window once live data exists.

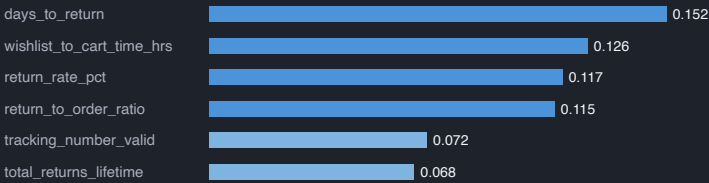
Random Forest vs. Logistic Regression on *return-fraud type*.

Spark MLlib's `GBTClassifier` only supports binary classification — not usable for this 4-class target — so Random Forest, which natively supports multi-class, is the primary model. Class-weighted training (inverse frequency) handles the 70/12/10/8 imbalance; evaluation reports per-class precision/recall/F1, not headline accuracy.

MODEL	WEIGHTED PRECISION	WEIGHTED RECALL	WEIGHTED F1
Logistic Regression (baseline)	0.998	0.998	0.998
Random Forest (final)	0.999	0.999	0.999

200 trees, maxDepth=10, class-weighted. Train 45,112 / test 14,888 rows, 75/25 split, seed=42.

TOP FEATURE IMPORTANCES



Engineered ratios rank alongside raw behavioral fields — the payoff of Phase 2.

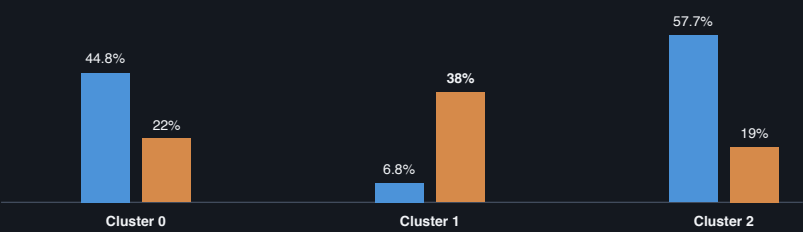
Honest note: this is a Kaggle synthetic dataset, so 0.999 F1 is higher than we'd expect in production — feature importance is spread across ~15 features with no single leak, but a realistic target on live data is **80–90% F1**. We treat this model as proof of the pipeline, not a performance guarantee. Full discussion in `docs/fraud-classification-results.md`.

APPROACH 2 — CLUSTERING

K-Means finds 3 behavioral segments *the fraud label can't see.*

An independent, unsupervised lens: features deliberately exclude the fraud-flag columns used by the classifier, so clustering surfaces behavior/quality patterns a supervised label alone can't. k=3 selected by silhouette score over k=2..6.

CLUSTER	N	AVG RETURN RATE	AVG ORDER VALUE	% QUALITY-ISSUE	DOMINANT MIX
0 — High-value fraud-risk	6,646	44.8%	\$484	22%	Fraudulent Return-heavy
1 — Core legitimate base	43,829	6.8%	\$157	38%	96% Legitimate
2 — Frequent policy-abuse	9,525	57.7%	\$129	19%	Policy Abuser-heavy



KEY INSIGHT

Cluster 1 has by far the **lowest** return rate but the **highest** quality-issue share — the two metrics move in opposite directions, which is exactly why a single blended "return rate" KPI hides this segment's problem from whoever owns it today.

This is a **QA/vendor problem**, not a fraud problem — cross-reference dominant product_category within Cluster 1 against return_reason to target the escalation.

Four interventions, four owners — *ship the free one first.*

SHIP THIS FIRST, NO ML REQUIRED

Require photo evidence above a value threshold. The "high-value item + no photo evidence" pattern alone touches **\$1.88M** of non-legitimate refund exposure across **4,673 orders** — a policy rule, live immediately, independent of the classifier's rollout timeline.

01 Deploy the fraud classifier as a triage score, not an auto-block.

Route the highest `fraud_signal_score` decile + Cluster 0 to manual review; auto-approve the lowest-risk tail. **Re-threshold asymmetrically** — a false "Legitimate → Fraudulent Return" flag costs more in trust than a slow review costs in ops time.

OWNER · FRAUD & RISK

02 Give Cluster 1's 38% quality-issue rate to QA as a systemic audit, not a vendor hunt.

Quality-issue rates are broad-based — banded 31.5%–35.2% across all 12 categories, not one bad vendor. **73% of customers**, near-zero fraud, but the highest quality complaint rate of any segment.

OWNER · QA / VENDOR MANAGEMENT

03 Tighten policy — don't investigate — for Cluster 2.

High order/return volume, low order value, Policy-Abuser-dominated. A shorter return window or restocking fee targets this pattern more precisely than a fraud investigation would.

OWNER · CUSTOMER POLICY

04 Feed the refund-exposure regressor to Finance, not just Fraud & Risk.

Linear Regression prices expected refund dollars the moment a return is filed — a reserve-accounting input independent of whether it turns out to be fraud, abuse, or genuine.

OWNER · FINANCE / CUSTOMER OPS



THANK YOU

Thank you. Questions *welcome.*

We'd rather be asked about the caveats than the number —
\$4.64M of non-legitimate refund exposure, **\$1.88M** of it
reachable today with a single no-ML policy rule.

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Dev Vagrecha · A049

Samruddhi Pradhan · A054

Yatharth Vij · A072

IN THE REPOSITORY

notebooks/ — two executable
PySpark notebooks, Bronze to
models

WRITTEN UP

docs/ — per-model results, each
with its own weaknesses section

THE DELIVERABLES

deliverables/ — this deck and the
full consulting report, HTML + PDF

REPRODUCIBILITY

seed=42 throughout · every number
here read from the notebooks' own
output

Dataset: `ecommerce_return_abuse_dataset.csv` (Kaggle) · references, AI-usage disclosure and team contributions in the consulting report's Appendix.